Preface

This collection of MATLAB modules is a selection of MATLAB based *engineering* design projects on reinforcement learning (RL), a machine learning technique (ML). The projects are clustered into independent, self-contained, modules. *Working knowledge of MATLAB is essential for mastering the design project modules*. The projects come from various modern and challenging engineering disciplines including drones, aircraft, satellites, robots, vehicles, fuel cells, smart electric power systems, wind turbines, computer and communication networks, bioengineering, to name a few.

These MATLAB modules will enhance the corresponding course curricula of various RL courses taught worldwide in both graduate and undergraduate engineering programs so that students, class instructors, and engineers will benefit from them by supplementing their theoretical knowledge with practical engineering design applications.

MATLAB's Live Editor environment is very convenient for implementation of the RL mechanism, whose main components can be implemented using MATLAB live scripts and live functions combined with Simulink. Namely, the control agent can be implemented using a MATLAB live script program. The state of the system obtained via a feedback channel measurement, can be represented by a MATLAB live function. The reward (cost, performance) estimation via the upper feedback loop can be also calculated using a MATLAB live function. The environment (engineering system subjected to RL) can be modeled using Simulink (or in simpler cases using a live function). As a matter of fact, the *MATLAB live scripts interactive capabilities in conjunction with Simulink modeling and simulation capabilities can create an excellent, software oriented testbed for verification of almost all RL algorithms for various engineering systems*.

The user of these modules will learn what is specific to the engineering approach to RL and what distinguishes it from the corresponding computer science approach to RL. Each module has a very brief introduction into the theoretical results of considered RL techniques. After mastering design projects, the reader will feel confident to apply the presented RL techniques to various practical engineering problems.

The modules are written, and the projects designed, from the perspective and prospective of engineering feedback control systems, more specifically optimal feedback control of dynamic systems that evolve either in continuous- or discrete-time domains with emphasis on deterministic problem formulations over stochastic problem formulations. *Bellman's dynamic programming methodology in continuous- and discrete-time domains* (which is a synonym for optimal feedback control in continuous- and discrete-time) *forms the mathematical foundation of these modules*.

The modules provide full coverage of many important results within the engineering approach to RL, which is much more general than the computer science approach to RL that is heavily limited by the imposed structure of Markov Decision Processes, to which many engineering systems (real word dynamic systems) hardly obey.

In addition to the RL design projects, the modules contains several exercises and homework problems done using MATLAB and Simulink for real physical engineering systems, which can serve as an excellent preparation for the corresponding engineering graduate and undergraduate courses on RL.

The complete information regarding to the modules, including project descriptions, exercises, homework, and some solutions will be posted on Professor Zoran Gajic website at Rutgers University https://sites.rutgers.edu/zoran-gajic, and for a public use on GitHub at github.com/ControlSystems-Rutgers/RL-for-Engineering-MATLAB-Design-Projects, and eventually on MathWorks website.

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